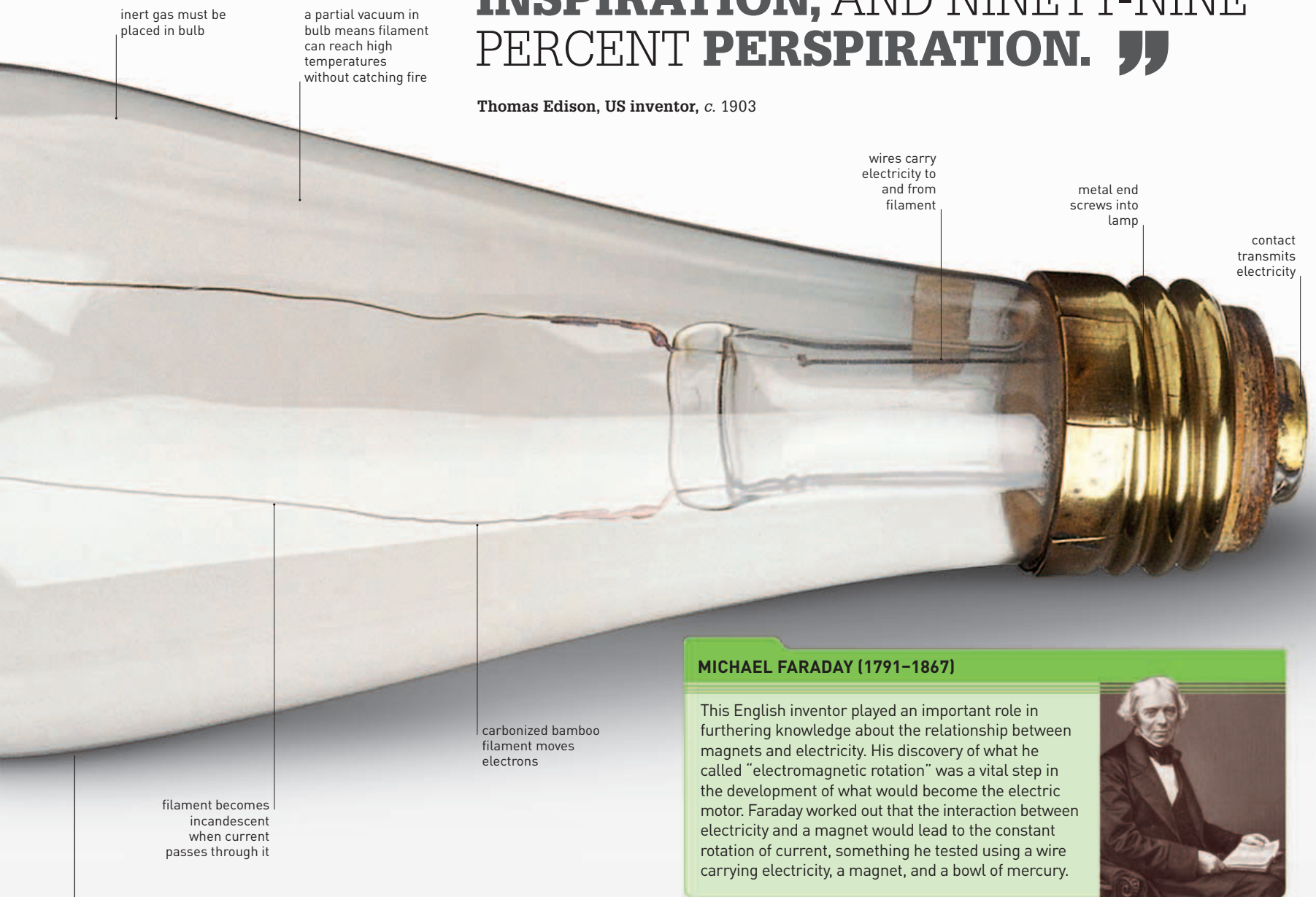


“ GENIUS IS ONLY ONE PERCENT INSPIRATION, AND NINETY-NINE PERCENT PERSPIRATION. ”

Thomas Edison, US inventor, c. 1903



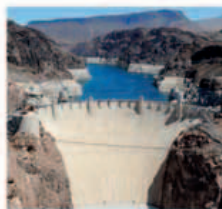
MICHAEL FARADAY (1791–1867)

This English inventor played an important role in furthering knowledge about the relationship between magnets and electricity. His discovery of what he called “electromagnetic rotation” was a vital step in the development of what would become the electric motor. Faraday worked out that the interaction between electricity and a magnet would lead to the constant rotation of current, something he tested using a wire carrying electricity, a magnet, and a bowl of mercury.



1878–79

The electric light bulb
British inventor Joseph Swan creates an incandescent “electric lamp.” The idea is improved by US scientist Thomas Edison, and the light bulb is born.



Hoover Dam

1882

Hydroelectric power
Scientists begin to realize that the force of water can generate electricity, and build dams and hydroelectric power plants to harness this energy.



Steam turbine

1884

Electricity from steam
Like liquid water, steam is also harnessed for electricity by devices like the turbine, created by Charles Patton.

Early 21st century

Sustainable electricity
Growing worries about pollution caused by the older ways of generating energy lead to the development of “green” technology, such as wind turbines.

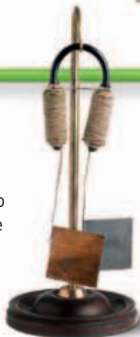


Wind turbines

1825

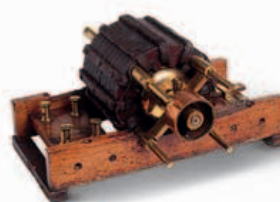
The electromagnet
This device, built by Joseph Henry, uses two metal plates, which are put in acid in order to form a voltaic cell.

Henry's electromagnet



1881

The world's first public electric lighting
The English town of Godalming, Surrey, brings to a close the era of the gas lamp when it wires its streets with electric lighting.



Tesla coil

1883–84

The Tesla coil
Serbian-American inventor Nikola Tesla develops a coil that can transmit electricity over long distances—it is a crucial discovery that aids the spread of electricity.

1950s

Nuclear power
Scientists discover that atoms can be used—in controlled nuclear reactions—as a source of energy to heat water, which then generates electricity. By 1951, the first nuclear power plant is built in Arco, Idaho, in the US.